

POLARIZATION

1. BASIC SETUP

The following is a natural way of extending a distance function on a graph:

Lemma (Extending to a Metric): Let $G = (V, E)$ be a (not necessarily finite) connected, simple graph and $d : E \rightarrow \mathbb{R}_+$ be a distance function satisfying the triangle inequality, that is, for all $u, v, w \in V$ such that $(u, v), (v, w), (u, w) \in E$ it holds

$$d(u, w) \leq d(u, v) + d(v, w)$$

Then we can extend d to a metric $\bar{d}$ on X by using the length of shortest paths.

Now we can define what intuitively means by “adding the points on the edges” to a graph:

Definition (Metric Graph): Given a graph $G = (V, E)$ and a function $d : E \rightarrow \mathbb{R}_+$, we can extend d to a metric on V .

The set of metric vertices as the union of *graph vertices* and *edge vetices*, the latter of which are denoted as symbolic pairs $\langle e, \lambda \rangle$:

$$\mathcal{V} := V \cup \{ \langle e, \lambda \rangle \mid e \in E, 0 < \lambda < d(e) \}$$

and define the map $\bar{d}$ via

$$\begin{aligned} \bar{d}(v, w) &:= d(v, w) & \bar{d}((v, w)_\lambda, v) &:= \lambda \\ \bar{d}(\langle e, \lambda \rangle, \langle e, \lambda' \rangle) &:= |\lambda - \lambda'| & \bar{d}((v, w)_\lambda, w) &:= d(e) - \lambda \end{aligned}$$

which can again be extended to a metric. We call $\mathcal{G} := (\mathcal{V}, \bar{d})$ the *metric graph* induced by G and d . For simplicity, we will often refer to the metric graph as G and the metric as d .

The metric graph is a metric space, thus we can consider continuous functions on it. A path in a metric graph can now be viewed under two different perspectives: As a finite sequence of vertices and edges, or as a continuous curve in the metric space. We will use both perspectives, depending on the context.

Definition (Kernel): A *kernel* is a map $K : [0, \infty) \rightarrow (-\infty, \infty]$ s.t:

1. $K(0) = \infty$
2. K is real-valued, strictly decreasing, and convex on $(0, \infty)$

Examples for kernels are $K(x) = \frac{1}{x^s}$ for $s > 0$ and $K(x) = \log(\frac{1}{x})$. Clearly kernels are closed under multiplication with positive scalars and under additions with decreasing convex functions. The ladder includes constants, other kernels, and left-translated kernels, that is, functions of the form $x \mapsto K(x + t)$ for some $t \geq 0$.

Definition (Sum of Translates): Let G be a metric graph, $p \in \mathbb{N}$ and let K be a kernel. Given $x \in \mathcal{V}^p$ and $t \in \mathcal{V}$ define the *pure sum of translates* as

$$f(x, t) := \sum_{i=1}^p K(d(t, x_i))$$

Additionally given a functions $J : \mathcal{V} \rightarrow (-\infty, \infty]$ which is real-valued at at least p points, we can define the *weighted sum of translates*

$$F(x, t) := J(t) + \sum_{i=1}^p K(d(t, x_i)) = J(t) + f(x, t)$$

Definition (Polarization Problem): Let $\mathcal{X} \subseteq \mathcal{V}^p$ be a set of feasible center placements and $x = (x_1, \dots, x_p) \in \mathcal{X}$ be a vector in that set. Denote with $X := \{x_1, \dots, x_p\}$ the corresponding set, and the connected components of $\mathcal{V} \setminus X$ as $\mathcal{V}_1(x), \dots, \mathcal{V}_k(x)$.

The *valleys* of F are the minima on the connected components:

$$m_j(x) := \min_{t \in \mathcal{V}_j(x)} F(x, t) \quad \text{for } j = 1, \dots, k$$

We further define

$$\underline{m}(x) := \min_{1 \leq j \leq k} m_j(x) = \inf_{t \in \mathcal{V}} F(x, t)$$

$$\overline{m}(x) := \max_{1 \leq j \leq k} m_j(x)$$

We call j *active* if $m_j(x) = \underline{m}(x)$, that is, if the minimum is attained on $\mathcal{V}_j$. Clearly $\underline{m}(x) \leq \overline{m}(x)$ and equality occurs if and only if all j are active. If equality occurs, we call x an *equioscillation point*. Now we can define the problems we are interested in:

$$m(\mathcal{X}) := \sup_{x \in \mathcal{X}} \underline{m}(x) = \sup_x \inf_j m_j(x) \quad (\text{Polarization problem})$$

$$M(\mathcal{X}) := \inf_{x \in \mathcal{X}} \overline{m}(x) = \inf_x \sup_j m_j(x) \quad (\text{dual Polarization problem})$$

We are mainly interested in computing the polarization problem for $X = \mathcal{V}^p$, but it is also interesting to consider the problem for smaller sets, especially finite sets, like for example V .

- The valleys don't have to be attained at a single point, but can be attained on a whole set.

2. THE LINE CASE

In this chapter we assume the simplest case: G only consists of two vertices connected by an edge, that is, homeomorphic to the unit interval $[0,1]$. Thus we can identify $\mathcal{V}$ with the unit interval and d with the usual euclidean distance. In this case, the connected components of $\mathcal{V} \setminus \{x_1, \dots, x_p\}$ are just the intervals between the points x_i and the endpoints 0 and 1.

We can formulate the problem as the following nonlinear program (P):

$$\begin{aligned} \max \quad & m \\ \text{s.t.} \quad & m \leq F(x, t) \text{ for all } t \in [0, 1] \\ & m \in \mathbb{R} \\ & x \in [0, 1]^p \end{aligned}$$

The following strong optimality criterion was shown (under stronger assumptions) in [1] and later generalized in [2].

Theorem 2.1 (Optimality criterion for Polarization on the Line): let K_i be a kernels. Then the following are equivalent for $x \in [0, 1]^p$:

1. x is optimal for the polarization problem
2. x is optimal for the dual polarization problem
3. x is an equioscillation point

The optimizer is unique up to permutation; equivalently unique after imposing $x_1 < \dots < x_p$. It fulfills $x_i \neq x_{i'}$ for all $i \neq i'$.

3. TREES

In the following we assume that we are given a finite metric tree G . Furthermore we assume that the kernel functions $K_i = K$ are identical.

3.1. Continuity of Component Minima.

To prove equioscillation necessity via perturbation arguments, we first establish continuity of the valley heights as functions of center position. This is the foundation for the contradiction proof.

Definition (Regular Center Configuration): Let $x = (x_1, \dots, x_p) \in \mathcal{V}^p$ be a placement of p centers on the metric tree G . We call x a *regular configuration* if:

1. No center coincides with a graph vertex: $x_i \in \mathcal{E}$ (interior of edges) for all i .
2. The removal of x yields exactly k connected components $\mathcal{V}_1, \dots, \mathcal{V}_k$ with $k \geq 2$ (no singleton components after removal).
3. Each component contains at least one graph vertex (non-degenerate Voronoi regions).

The set of all regular configurations on $\mathcal{V}$ is denoted $\mathcal{X}_{\text{reg}} \subseteq \mathcal{V}^p$.

Lemma (Minimum over a Fixed Compact Set is Continuous): Let X be a metric space, T a compact metric space, and $f : X \times T \rightarrow \mathbb{R}$ continuous. Define

$$v(x) := \min_{t \in T} f(x, t).$$

Then $v : X \rightarrow \mathbb{R}$ is continuous.

Proof: Let $x_n \rightarrow x$. Choose $t_n \in T$ with $v(x_n) = f(x_n, t_n)$. Since T is compact, after passing to a subsequence we may assume $t_n \rightarrow t_* \in T$. By continuity of f ,

$$\liminf_{n \rightarrow \infty} v(x_n) = \liminf_{n \rightarrow \infty} f(x_n, t_n) = f(x, t_*) \geq v(x).$$

On the other hand, for any fixed minimizer $\hat{t} \in \arg \min_{t \in T} f(x, t)$,

$$\limsup_{n \rightarrow \infty} v(x_n) \leq \limsup_{n \rightarrow \infty} f(x_n, \hat{t}) = f(x, \hat{t}) = v(x).$$

Hence $v(x_n) \rightarrow v(x)$, so v is continuous. $\square$

Lemma (Continuity of Valley Heights under Regular Placement): Let $x^0 \in \mathcal{X}_{\text{reg}}$ and fix a component index j . Then there exist a neighborhood U of x^0 in $\mathcal{X}_{\text{reg}}$ and a compact set $C_j \subseteq \mathcal{V}$ such that for all $x \in U$:

1. $C_j \subseteq \mathcal{V}_{j(x)}$,
2. $\min_{t \in \mathcal{V}_{j(x)}} F(x, t) = \min_{t \in C_j} F(x, t)$.

Consequently, $m_{j(x)} := \min_{t \in \mathcal{V}_{j(x)}} F(x, t)$ is continuous on U and therefore on $\mathcal{X}_{\text{reg}}$.

Proof: Fix $x^0 \in \mathcal{X}_{\text{reg}}$ and j .

Because x^0 is regular, all centers are in edge interiors and the component decomposition is locally stable: there exists a neighborhood U_0 of x^0 in which the same component labels persist and each $\mathcal{V}_{j(x)}$ differs from $\mathcal{V}_{j(x^0)}$ only by small boundary slivers near the moving centers.

Since $K(r) \rightarrow \infty$ as $r \downarrow 0$, points sufficiently close to any center have arbitrarily large potential value. Choose $M > m_{j(x^0)} + 1$. By continuity of F , after shrinking to a neighborhood $U \subseteq U_0$ we can ensure:

1. every point in those boundary slivers has value $F(x, t) > M$ for all $x \in U$,
2. there exists a nonempty compact interior core $C_j \subseteq \mathcal{V}_{j(x^0)}$ with $F(x, t) \leq M - 1$ for some $t \in C_j$ and all $x \in U$.

Therefore no minimizer of $F(x, \cdot)$ on $\mathcal{V}_{j(x)}$ can lie in the slivers, so every minimizer lies in the common compact set C_j . Hence

$$m_{j(x)} = \min_{t \in \mathcal{V}_{j(x)}} F(x, t) = \min_{t \in C_j} F(x, t)$$

for all $x \in U$.

Now apply the previous theorem to the continuous map $(x, t) \mapsto F(x, t)$ on $U \times C_j$ with fixed compact feasible set C_j . It follows that $x \mapsto \min_{t \in C_j} F(x, t)$ is continuous on U , hence m_j is continuous on U .

Since x^0 and j were arbitrary, m_j is continuous on all of $\mathcal{X}_{\text{reg}}$. $\square$ $\square$

Corollary 3.1.1: In the regular regime, continuity of valley heights can be reduced to a fixed-compact-set minimization problem. No global hemicontinuity argument for the full moving component correspondence is needed.

REFERENCES

1. Fenton, P.C.: A min-max theorem for sums of translates. *Journal of Mathematical Analysis and Applications*. (2000)
2. Farkas, B., Nagy, B., Révész, S.: On the weighted Bojanov–Chebyshev problem and the sum of translates method of Fenton. *Sbornik: Mathematics*. 214, 1163–1190 (2023)